

Final Project

Dee Christen

2026-05-06

#Final Project For my final project I looked at a F1 strategy data set.

```
df <- read.csv("f1_strategy_dataset_v4.csv")
```

First I looked to see if I needed to clean the data.

```
head(df)
```

```
##   Driver LapNumber Compound Stint TyreLife Position LapTime..s.
## 1   ALB          1   MEDIUM    1        2         17    100.625
## 2   ALB          2   MEDIUM    1        3         18     93.560
## 3   ALB          3   MEDIUM    1        4         18     91.768
## 4   ALB          4   MEDIUM    1        5         18     91.591
## 5   ALB          5   MEDIUM    1        6         18     91.422
## 6   ALB          6   MEDIUM    1        7         18     91.491
##
##           Race Year LapTime_Delta Cumulative_Degradation PitStop
## 1 Abu Dhabi Grand Prix 2023          0.000              0.000      0
## 2 Abu Dhabi Grand Prix 2023         -7.065             -7.065      0
## 3 Abu Dhabi Grand Prix 2023         -1.792             -8.857      0
## 4 Abu Dhabi Grand Prix 2023         -0.177             -9.034      0
## 5 Abu Dhabi Grand Prix 2023         -0.169             -9.203      0
## 6 Abu Dhabi Grand Prix 2023          0.069             -9.134      0
##   PitNextLap RaceProgress Normalized_TyreLife Position_Change
## 1           0   0.01724138          0.1176471              0
## 2           0   0.03448276          0.1764706             -1
## 3           0   0.05172414          0.2352941              0
## 4           0   0.06896552          0.2941176              0
## 5           0   0.08620690          0.3529412              0
## 6           0   0.10344828          0.4117647              0
```

There is no null values in the data.

```
sum(is.na(df))
```

```
## [1] 0
```

Next, I wanted to see if there were any duplicates or strange data. There was not any in the driver or race column.

```
unique(df$Driver)
```

```
## [1] "ALB" "ALO" "BOT" "GAS" "HAM" "HUL" "LEC" "MAG" "NOR" "OCO" "PER" "PIA"
## [13] "RIC" "RUS" "SAI" "SAR" "STR" "TSU" "VER" "ZHO" "DEV" "LAW" "COL" "DOO"
## [25] "BEA" "ANT" "BOR" "HAD" "LAT" "MSC" "VET"
```

```
unique(df$Race)
```

```
## [1] "Abu Dhabi Grand Prix"      "Australian Grand Prix"
## [3] "Austrian Grand Prix"       "Azerbaijan Grand Prix"
## [5] "Bahrain Grand Prix"       "Belgian Grand Prix"
## [7] "British Grand Prix"       "Canadian Grand Prix"
## [9] "Dutch Grand Prix"         "Hungarian Grand Prix"
## [11] "Italian Grand Prix"       "Japanese Grand Prix"
## [13] "Las Vegas Grand Prix"     "Mexico City Grand Prix"
## [15] "Miami Grand Prix"        "Monaco Grand Prix"
## [17] "Pre-Season Testing"      "Qatar Grand Prix"
## [19] "Saudi Arabian Grand Prix" "Singapore Grand Prix"
## [21] "Spanish Grand Prix"      "São Paulo Grand Prix"
## [23] "United States Grand Prix" "Chinese Grand Prix"
## [25] "Emilia Romagna Grand Prix" "French Grand Prix"
## [27] "Pre-Season Test"        "Pre-Season Track Session"
```

In the tire compound column, there were values listed as none, which doesn't make sense because you need tires on your car to race. In doing more research, I found there were 66 rows that had none.

```
unique(df$Compound)
```

```
## [1] "MEDIUM"      "HARD"      "SOFT"      "INTERMEDIATE" "WET"
## [6] "None"
```

```
sum(df$Compound == "None")
```

```
## [1] 66
```

I looked into the data where there were none to see if there were actual rows or maybe the data had been entered incorrectly. I looked at the data and found that it should be hard tires. I fixed the issue by mutating the rows.

```
filter(df, Compound == "None")
```

```
df <- df %>%
  mutate(Compound = recode(Compound, "None" = "HARD"))
```

```
unique(df$Compound)
```

```
## [1] "MEDIUM"      "HARD"      "SOFT"      "INTERMEDIATE" "WET"
```

```
sum(df$Compound == "None")
```

```
## [1] 0
```

When I was looking at the races, I noticed that it had the pre-season testing in the data set as well. For this analysis, I just wanted the Grand Prix data. I created a new data frame with just the Grand Prix data.

```
unique(df$Race)
```

```
## [1] "Abu Dhabi Grand Prix"      "Australian Grand Prix"
## [3] "Austrian Grand Prix"       "Azerbaijan Grand Prix"
## [5] "Bahrain Grand Prix"       "Belgian Grand Prix"
## [7] "British Grand Prix"       "Canadian Grand Prix"
## [9] "Dutch Grand Prix"         "Hungarian Grand Prix"
## [11] "Italian Grand Prix"       "Japanese Grand Prix"
## [13] "Las Vegas Grand Prix"     "Mexico City Grand Prix"
## [15] "Miami Grand Prix"        "Monaco Grand Prix"
## [17] "Pre-Season Testing"      "Qatar Grand Prix"
## [19] "Saudi Arabian Grand Prix" "Singapore Grand Prix"
## [21] "Spanish Grand Prix"      "São Paulo Grand Prix"
```

```
## [23] "United States Grand Prix" "Chinese Grand Prix"
## [25] "Emilia Romagna Grand Prix" "French Grand Prix"
## [27] "Pre-Season Test"          "Pre-Season Track Session"
```

```
rdf <- df %>%
  filter(Race != "Pre-Season Test") %>%
  filter(Race != "Pre-Season Track Session") %>%
  filter(Race != "Pre-Season Testing")
```

```
unique(rdf$Race)
```

```
## [1] "Abu Dhabi Grand Prix"      "Australian Grand Prix"
## [3] "Austrian Grand Prix"       "Azerbaijan Grand Prix"
## [5] "Bahrain Grand Prix"        "Belgian Grand Prix"
## [7] "British Grand Prix"        "Canadian Grand Prix"
## [9] "Dutch Grand Prix"          "Hungarian Grand Prix"
## [11] "Italian Grand Prix"         "Japanese Grand Prix"
## [13] "Las Vegas Grand Prix"       "Mexico City Grand Prix"
## [15] "Miami Grand Prix"           "Monaco Grand Prix"
## [17] "Qatar Grand Prix"           "Saudi Arabian Grand Prix"
## [19] "Singapore Grand Prix"       "Spanish Grand Prix"
## [21] "São Paulo Grand Prix"       "United States Grand Prix"
## [23] "Chinese Grand Prix"         "Emilia Romagna Grand Prix"
## [25] "French Grand Prix"
```

I noticed that the lap time column name was weirdly named, so I re-named it LapTime.

```
rdf <- rdf %>%
  rename(LapTime = LapTime..s.)
```

I wanted to see the difference between the average lap time for just the Grand Prix race and the Grand Prix plus pre-season testing.

```
mean(rdf$LapTime)
```

```
## [1] 92.19148
```

```
mean(df$LapTime..s.)
```

```
## [1] 92.58719
```

I wanted to see who had the fastest lap. It was Max Verstappen. I also noticed that all of the fastest laps were at the Austrian Grand Prix.

```
arrange(rdf, LapTime) %>% slice(1:10)
```

```
##   Driver LapNumber Compound Stint TyreLife Position LapTime
## 1   VER          71    SOFT     5         5         1  67.012
## 2   LEC          62   MEDIUM  4         4         1  67.583
## 3   ALO          70    SOFT     4         2        18  67.694
## 4   VER          68    SOFT     4         7         5  67.719
## 5   LEC          63   MEDIUM  4         5         1  67.814
## 6   PIA          59   MEDIUM  3         6         2  67.924
## 7   NOR          53   MEDIUM  3         2         2  68.016
## 8   LEC          61   MEDIUM  4         3         1  68.070
## 9   VER          70    SOFT     4         9         5  68.072
## 10  PIA          60   MEDIUM  3         7         2  68.085
##                                     Race Year LapTime_Delta Cumulative_Degradation PitStop
## 1 Austrian Grand Prix 2023          -20.818              -20.818              0
```

```
## 2 Austrian Grand Prix 2022 -10.956 -53.764 1
## 3 Austrian Grand Prix 2024 -22.667 -47.349 0
## 4 Austrian Grand Prix 2024 -3.972 -47.292 0
## 5 Austrian Grand Prix 2022 -9.915 -53.533 1
## 6 Austrian Grand Prix 2025 -7.180 -38.033 1
## 7 Austrian Grand Prix 2024 -9.898 -40.116 1
## 8 Austrian Grand Prix 2022 -10.762 -53.277 1
## 9 Austrian Grand Prix 2024 -0.249 -46.939 0
## 10 Austrian Grand Prix 2025 -6.413 -37.872 1
## PitNextLap RaceProgress Normalized_TyreLife Position_Change
## 1 0 1.0000000 1.0000000 0
## 2 0 0.8611111 0.09302326 3
## 3 0 0.9210526 0.07407407 0
## 4 0 0.8717949 0.31818182 0
## 5 0 0.8750000 0.11627907 3
## 6 0 0.7564103 0.18181818 2
## 7 0 0.6794872 0.04878049 0
## 8 0 0.8472222 0.06976744 3
## 9 0 0.8974359 0.40909091 0
## 10 0 0.7692308 0.21212121 2
```

I looked at the average min and max for each Grand Prix to see the differences.

```
rdf %>%
  group_by(Race) %>%
  summarise(lapmean = mean(LapTime), lapmin = min(LapTime), lapmax = max(LapTime), lapmed = median(LapTime))
  arrange(lapmean)
```

```
## # A tibble: 25 x 5
## Race lapmean lapmin lapmax lapmed
## <chr> <dbl> <dbl> <dbl> <dbl>
## 1 Austrian Grand Prix 72.7 67.0 135. 71.1
## 2 Dutch Grand Prix 81.1 72.3 149. 76.8
## 3 São Paulo Grand Prix 82.1 72.4 151. 76.7
## 4 Canadian Grand Prix 83.7 74.1 150. 78.5
## 5 Spanish Grand Prix 84.2 75.7 149. 81.5
## 6 Hungarian Grand Prix 85.1 79.4 125. 84.3
## 7 Mexico City Grand Prix 85.4 78.3 149. 83.5
## 8 Emilia Romagna Grand Prix 86.2 78.0 150. 82.1
## 9 Italian Grand Prix 87.2 80.9 145. 86.1
## 10 Abu Dhabi Grand Prix 90.8 85.6 144. 89.9
## # i 15 more rows
```

I then wanted to see the average and min lap times for each driver.

```
rdf %>%
  group_by(Driver) %>%
  summarise(lapmean = mean(LapTime), lapmin = min(LapTime)) %>%
  arrange(lapmin)
```

```
## # A tibble: 31 x 3
## Driver lapmean lapmin
## <chr> <dbl> <dbl>
## 1 VER 91.4 67.0
## 2 LEC 91.7 67.6
## 3 ALO 92.3 67.7
```

```
## 4 PIA      90.4  67.9
## 5 NOR      91.6  68.0
## 6 PER      92.3  68.1
## 7 STR      92.3  68.5
## 8 HAM      91.7  68.6
## 9 SAI      91.2  68.9
## 10 GAS     92.8  69.0
## # i 21 more rows
```

I wanted to look at the average lap time for each tire compound.

```
rdf %>%
  group_by(Compound) %>%
  summarise(lapmean = mean(LapTime)) %>%
  arrange(lapmean)
```

```
## # A tibble: 5 x 2
##   Compound    lapmean
##   <chr>      <dbl>
## 1 SOFT      91.4
## 2 HARD      91.5
## 3 MEDIUM   91.6
## 4 INTERMEDIATE 102.
## 5 WET      107.
```

I wanted to look at how tire life plays into lap time so I created a new column.

```
rdf <- rdf %>%
  mutate(TireDeg = case_when(
    Compound == "SOFT" & TyreLife < 5 ~ "new",
    Compound == "SOFT" & TyreLife <= 15 ~ "used",
    Compound == "SOFT" & TyreLife > 15 ~ "old",

    Compound == "MEDIUM" & TyreLife < 10 ~ "new",
    Compound == "MEDIUM" & TyreLife <= 25 ~ "used",
    Compound == "MEDIUM" & TyreLife > 25 ~ "old",

    Compound == "HARD" & TyreLife < 15 ~ "new",
    Compound == "HARD" & TyreLife <= 30 ~ "used",
    Compound == "HARD" & TyreLife > 30 ~ "old",

    Compound == "INTERMEDIATE" & TyreLife < 10 ~ "new",
    Compound == "INTERMEDIATE" & TyreLife <= 30 ~ "used",
    Compound == "INTERMEDIATE" & TyreLife > 30 ~ "old",

    Compound == "WET" & TyreLife < 10 ~ "new",
    Compound == "WET" & TyreLife <= 30 ~ "used",
    Compound == "WET" & TyreLife > 30 ~ "old"
  ))
```

In looking at this data, you can see that newer tires get better lap times.

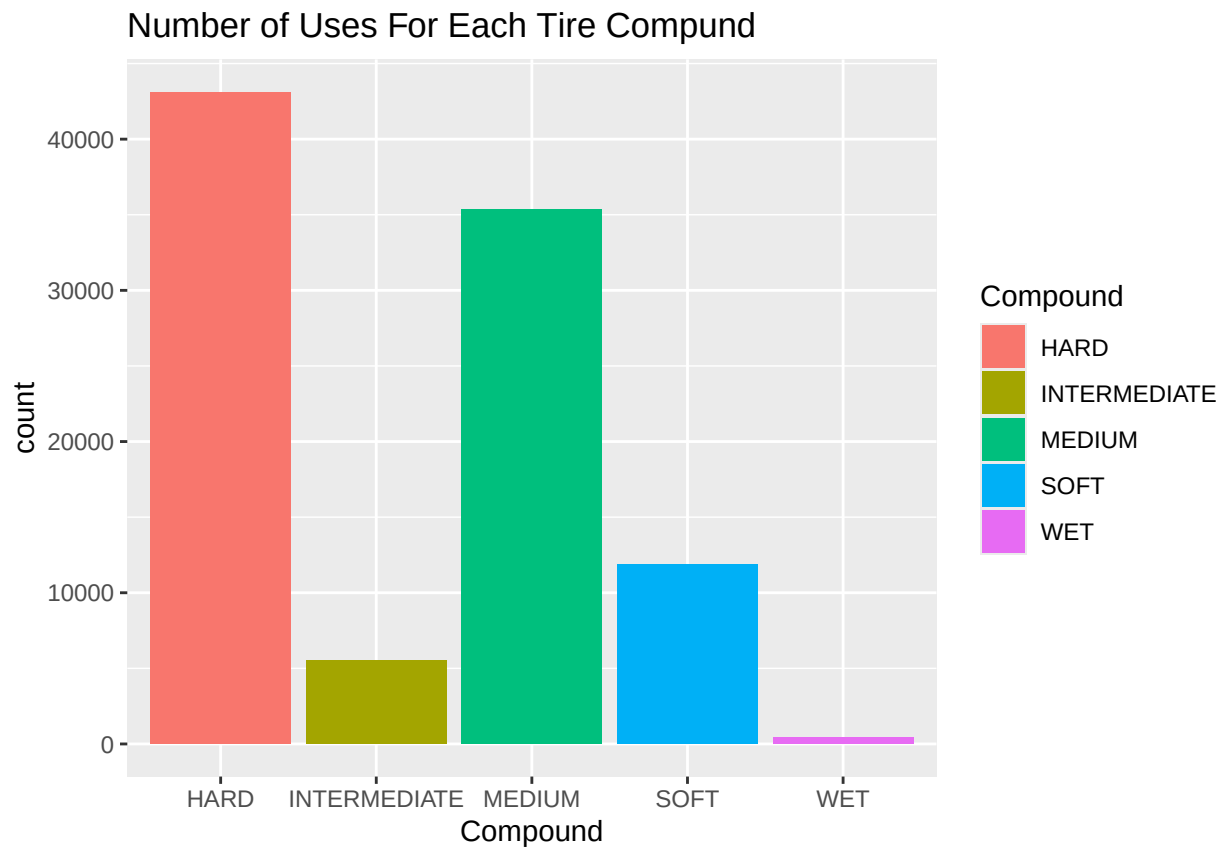
```
rdf %>%
  group_by(Compound, TireDeg) %>%
  summarise(lapmean = mean(LapTime), lapmin = min(LapTime)) %>%
  arrange(Compound, lapmin)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by Compound and TireDeg.
## i Output is grouped by Compound.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(Compound, TireDeg))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

## # A tibble: 14 x 4
## # Groups:   Compound [5]
##   Compound    TireDeg lapmean lapmin
##   <chr>      <chr>    <dbl>  <dbl>
## 1 HARD      new      94.0   68.1
## 2 HARD      used     89.4   69.0
## 3 HARD      old      88.0   69.5
## 4 INTERMEDIATE old     97.1   80.5
## 5 INTERMEDIATE used     97.3   80.5
## 6 INTERMEDIATE new     109.   81.2
## 7 MEDIUM    new      95.8   67.6
## 8 MEDIUM    used     89.2   68.5
## 9 MEDIUM    old      86.2   69.2
## 10 SOFT      used     90.8   67.0
## 11 SOFT      new      99.6   67.7
## 12 SOFT      old      86.9   70.5
## 13 WET       used     97.4   90.3
## 14 WET       new     114.   94.0
```

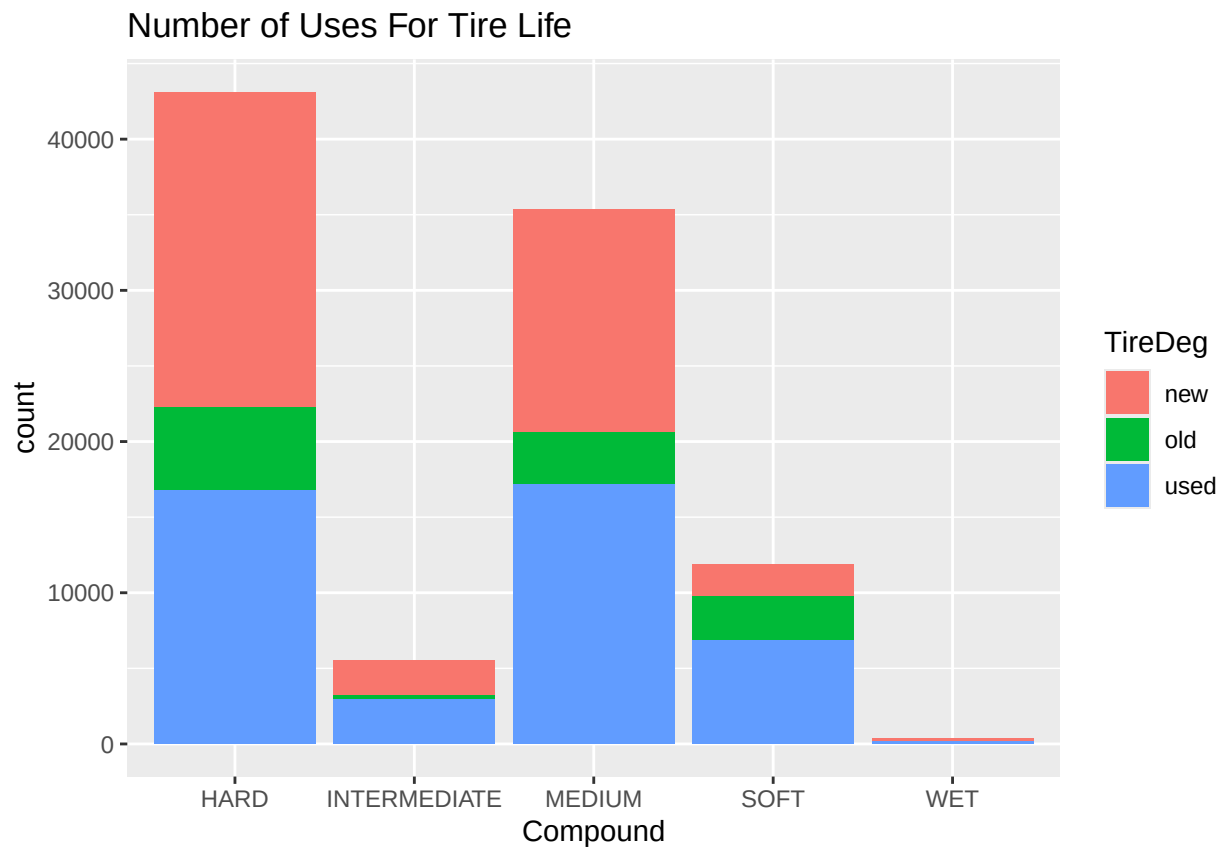
I wanted to see which tire was used most. Hard tires were used the most.

```
ggplot(data = rdf, aes(x = Compound, fill = Compound)) +
  geom_bar() +
  labs(title = "Number of Uses For Each Tire Compound")
```

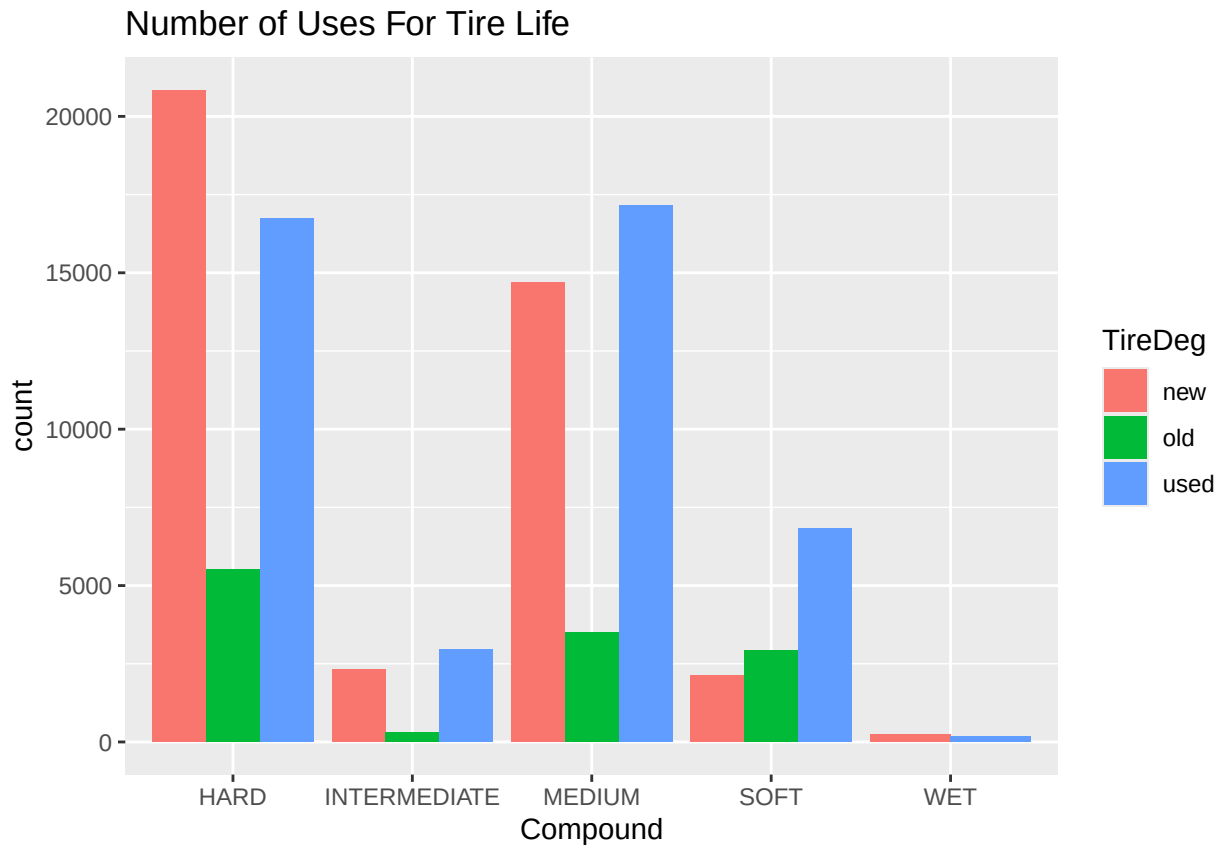


looked at the tire degradation.

```
ggplot(data = rdf, aes(x = Compound, fill = TireDeg)) +  
  geom_bar() +  
  labs(title = "Number of Uses For Tire Life")
```



```
ggplot(data = rdf, aes(x = Compound, fill = TireDeg)) +  
  geom_bar(position = "dodge") +  
  labs(title = "Number of Uses For Tire Life")
```

```

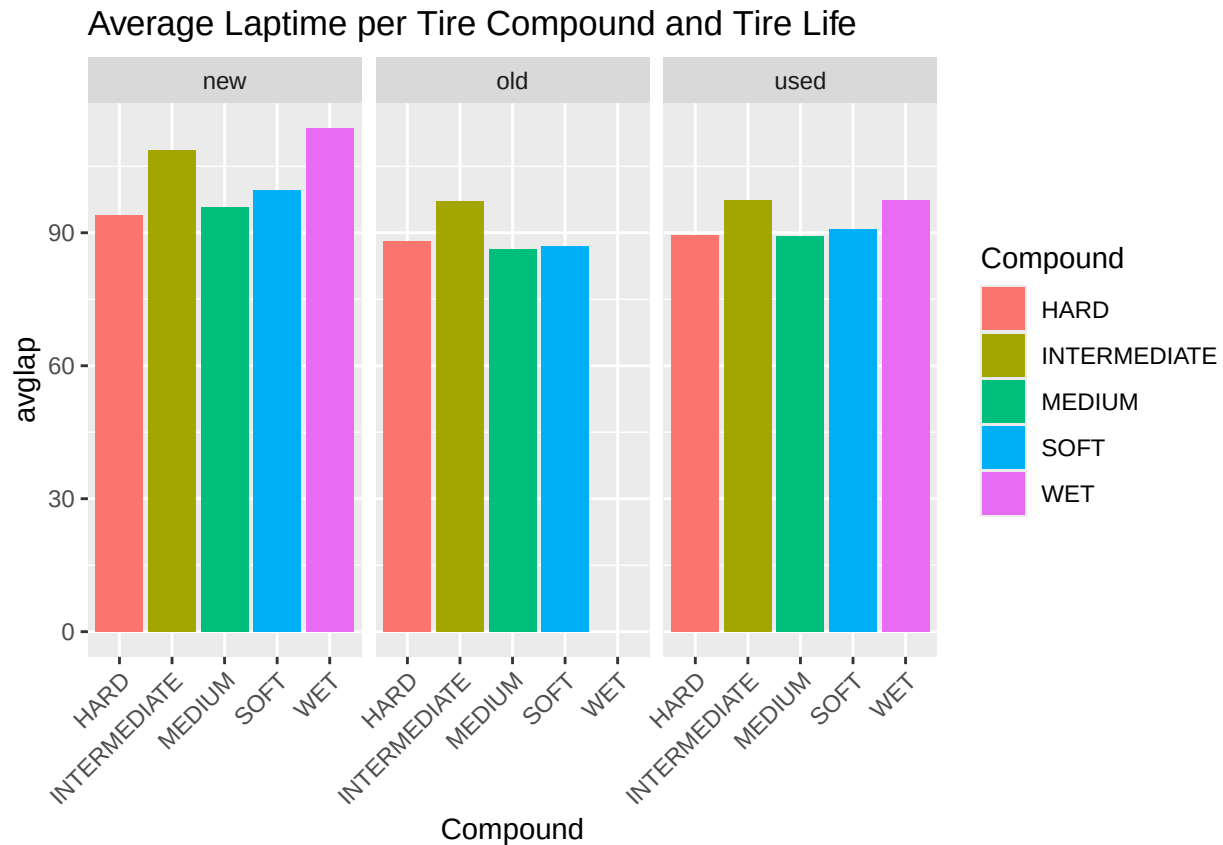
rdf %>%
  group_by(Compound, TireDeg) %>%
  summarise(avglap = mean(LapTime)) %>%
  ggplot(aes(x = Compound, y = avglap, fill = Compound)) +
  geom_col() +
  facet_wrap(~ TireDeg) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Average Laptime per Tire Compound and Tire Life")

```

```

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by Compound and TireDeg.
## i Output is grouped by Compound.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(Compound, TireDeg))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

```



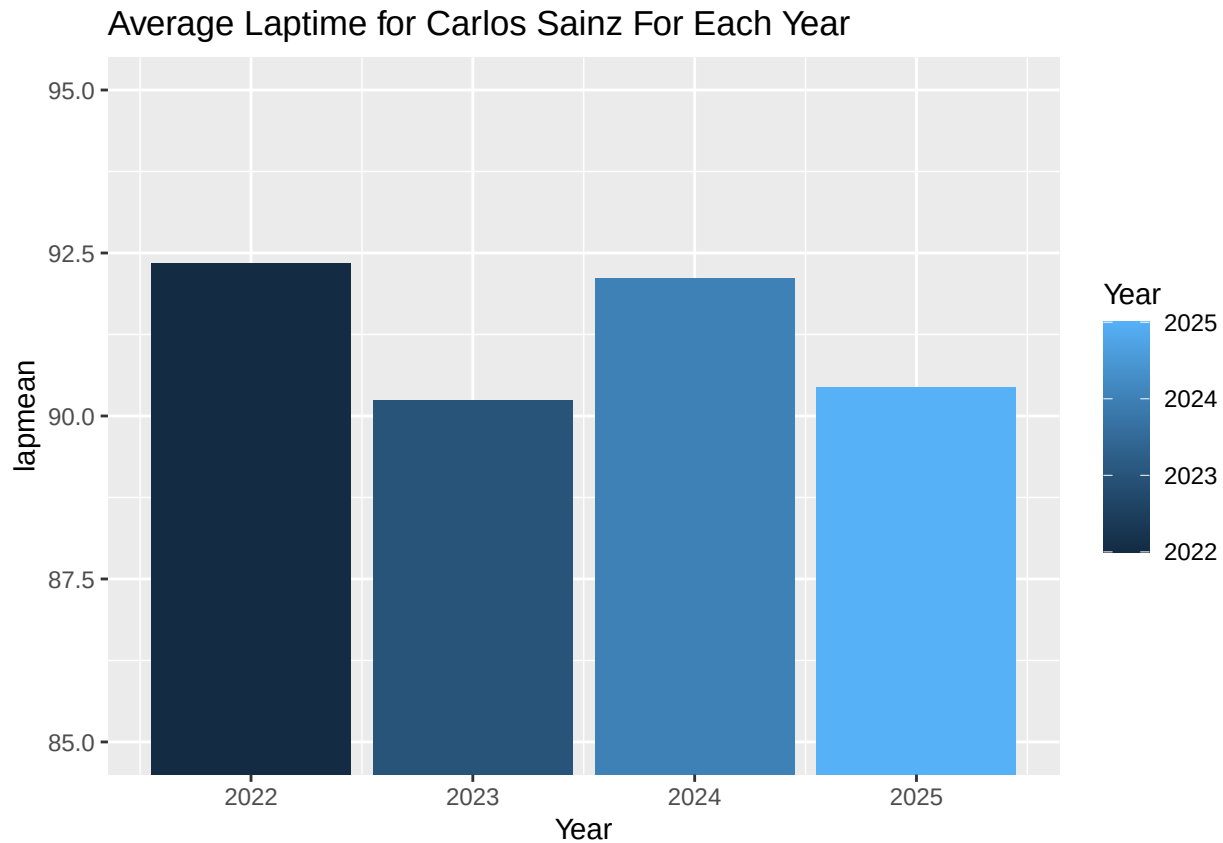
I created a new data frame to focus on Carlos Sainz to see the difference of lap times over the years.

```
sdf <- rdf %>%
  filter(Driver == "SAI")
```

I created a bar chart to see the difference of average lap times over the years.

```
summary_sdf <- sdf %>%
  group_by(Year) %>%
  summarise(lapmean = mean(LapTime))

ggplot(data = summary_sdf, aes(x = Year, y = lapmean, fill = Year)) +
  geom_col() +
  coord_cartesian(ylim = c(85, 95)) +
  labs(title = "Average Laptime for Carlos Sainz For Each Year")
```



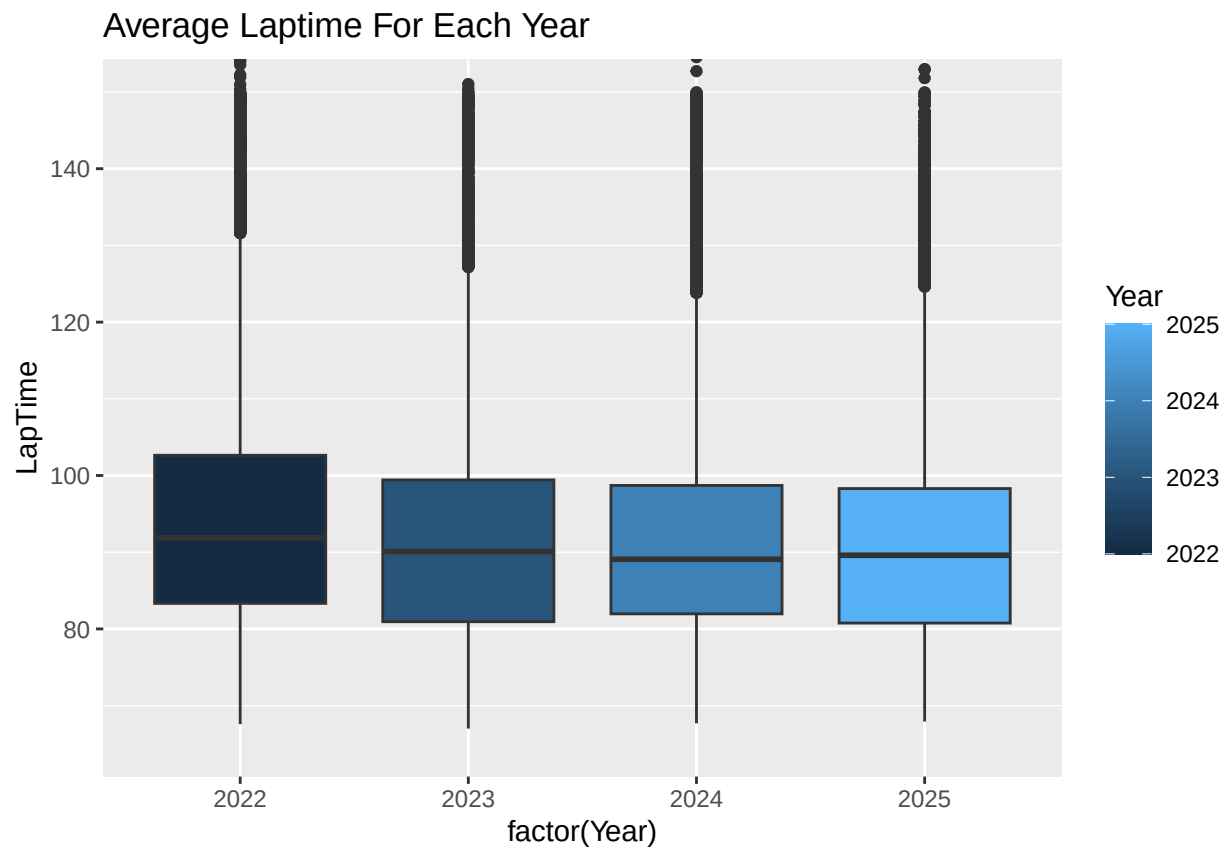
I chose this driver because he switched teams in 2025 and you can see that his fastest lap did go up in that year.

```
sdf %>%
  group_by(Year) %>%
  summarise(lapmean = mean(LapTime), lapmin = min(LapTime)) %>%
  arrange(lapmin)
```

```
## # A tibble: 4 x 3
##   Year lapmean lapmin
##   <int>   <dbl>   <dbl>
## 1  2023     90.2     68.9
## 2  2024     92.1     69.3
## 3  2022     92.3     69.4
## 4  2025     90.4     73.7
```

I created a box and whisker plot to chart the lap times over the years.

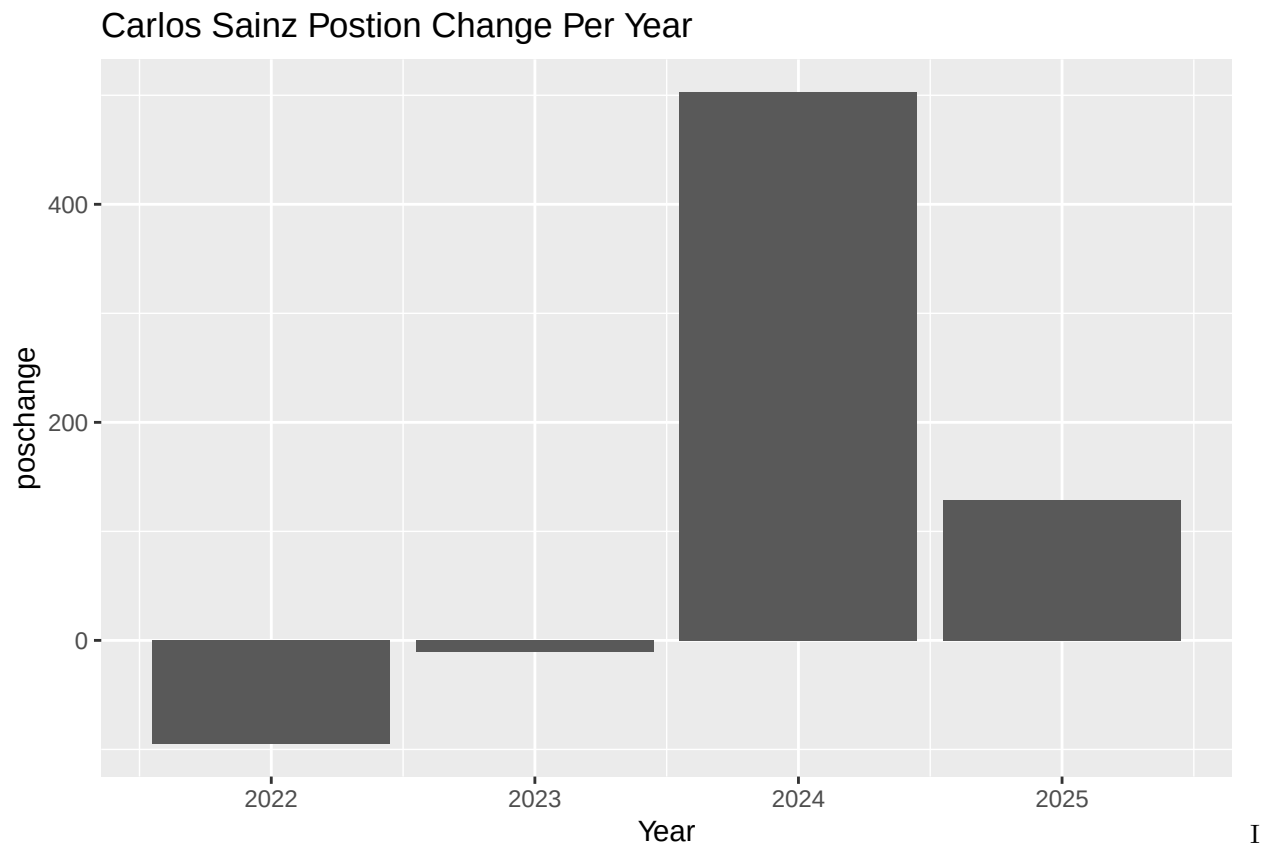
```
ggplot(rdf, aes(x = factor(Year), y = LapTime, fill = Year)) +
  geom_boxplot() +
  coord_cartesian(ylim = c(65, 150)) +
  labs(title = "Average Laptime For Each Year")
```



```
ssdf <- sdf %>%  
  group_by(Year) %>%  
  summarise(poschange = sum(Position_Change))
```

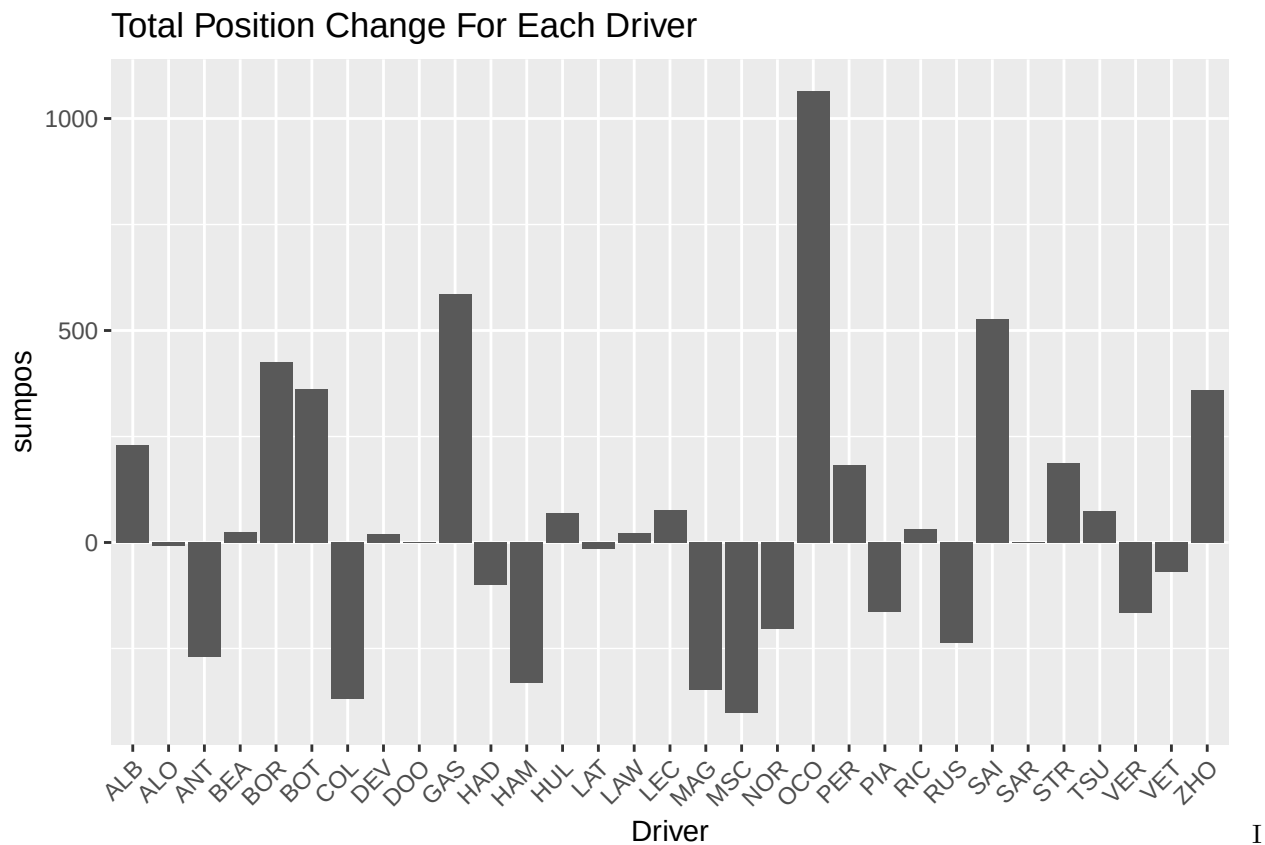
I looked at the position change for Carlos Sainz.

```
ggplot(data = ssdf, aes(x = Year, y = poschange)) +  
  geom_col() +  
  labs(title = "Carlos Sainz Postion Change Per Year")
```



then looked at the position change for each driver.

```
rdf %>%
  group_by(Driver) %>%
  summarise(sumpos = sum(Position_Change), .groups = "drop") %>%
  ggplot(aes(x = Driver, y = sumpos)) +
  geom_col() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Total Position Change For Each Driver")
```



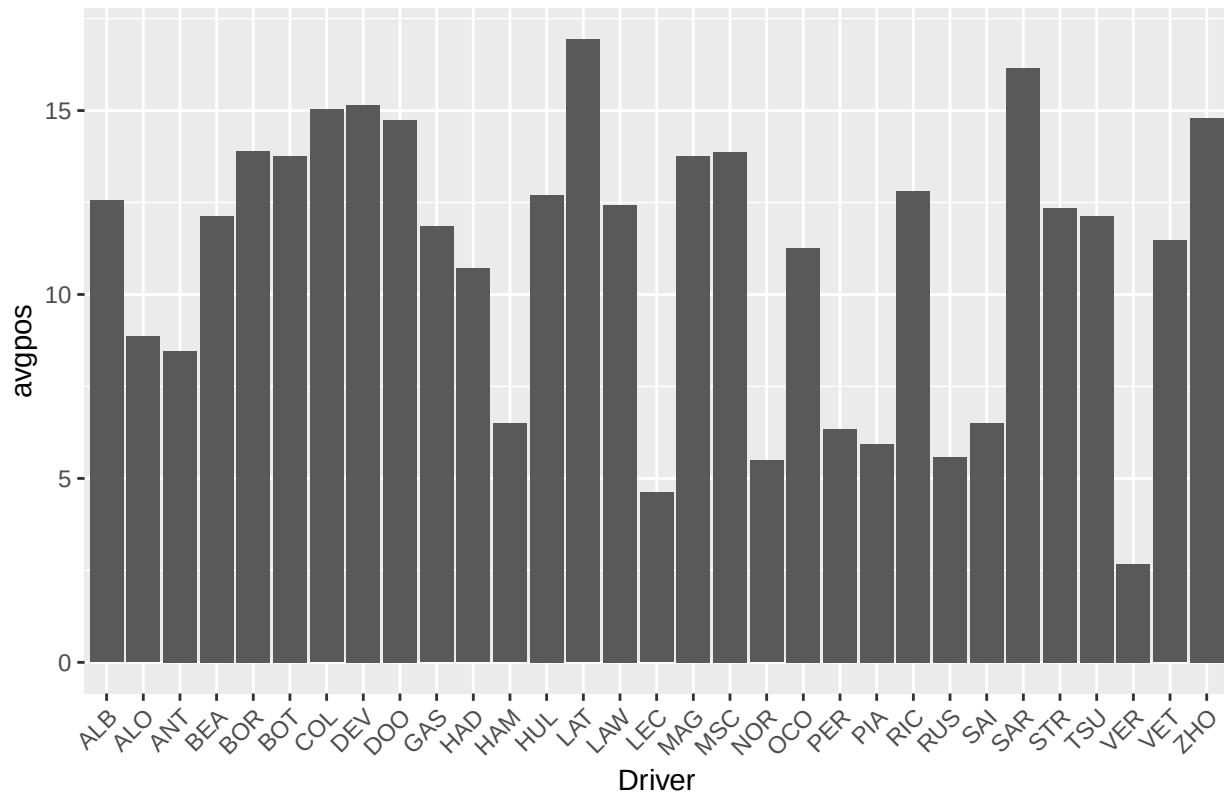
looked at the average driver position.

```

rdf %>%
  group_by(Driver) %>%
  summarise(avgpos = mean(Position, na.rm = TRUE)) %>%
  ggplot(aes(x = Driver, y = avgpos)) +
  geom_col() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Average position For Each Driver")

```

Average position For Each Driver



I

ran a t test to see if tire life and pitstop were related. My null hypothesis is they are related.

```
t.test(TyreLife ~ PitStop, data = rdf)
```

```
##
## Welch Two Sample t-test
##
## data: TyreLife by PitStop
## t = 76.07, df = 54602, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  4.890240 5.148908
## sample estimates:
## mean in group 0 mean in group 1
##      15.83341      10.81384
```

I reject my null hypothesis.